Michele Guida



Machine Learning Engineer

Deep Learning • LLMs / RAG • AI Security • Data Science

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Born 17 August 2002 • Italian citizen • Driving licence B

SUMMARY

Machine Learning Engineer at Exein building end-to-end AI/ML systems, from data pipelines and models to LLM/RAG applications. M.Sc. in Computer Engineering (Roma Tre University, 110/110 cum laude) and Visiting Research Scholar at the University of Alabama, with three IEEE papers accepted in 2026, two as first author. Strong in deep learning (PyTorch), multi-agent LLM systems and model evaluation, with an eval-first approach to reliable AI.

EXPERIENCE

Machine Learning Engineer : Exein

Nov 2026 – Present

Machine learning for embedded and IoT device security: on-device threat and anomaly detection.

Visiting Research Scholar : University of Alabama, Computer Science

Feb 2026 – Jul 2026

Awarded a competitive, merit-based scholarship to carry out my Master's thesis abroad (M.Sc. Computer Engineering, a.y. 2025/26): a joint Roma Tre and Alabama research exchange in applied machine learning.

Research projects:

- **Cognitive Firewall: Zero-Trust LLM-Safety Framework** — first author • ICMLA 2026 (accepted)
  - Cut jailbreak attack success to 2% or below on 3 of 4 benchmarks at 8% benign over-refusal, outperforming published guard models (Llama Guard, ShieldGemma).
  - Owned design, prototype and evaluation end-to-end: four zero-trust gates (intent, context, conversation, output) that block before generation with an auditable rationale; local LLMs via Ollama, LLM-as-judge scoring.

- **Agentra: Multi-Agent LLM System** — co-author • ICMLA 2026 (accepted)
  - Lifted decision F1 from 0.61 to 0.84 over a static baseline with a 0% projected harmful-action rate on 120 events.
  - Co-designed the system: a Planner proposes, independent Validators review in a bounded loop and a Moderator screens Graph RAG retrieval for prompt injection; actions are risk-scored and audited.

- **Shortcut Learning in Early Intrusion Detection** — M.Sc. thesis • first author • IEEE TPS 2026 (accepted)
  - Reached near full-flow accuracy from the first 4 to 5 packets with flow-statistics (Random Forest, XGBoost) and raw-byte models (BiGRU, Transformer) on 4 benchmarks (CIC-IDS2017/2018, ToN-IoT, CIC-IoT2023).
  - Showed these scores are inflated by dataset fingerprints: masking TCP header fields (options, window size) cut macro-F1 by up to 36 points; proposed generator-diversity and header-sensitivity checks.

PUBLICATIONS

- **M. Guida**, R. Shikhhamzayev, S. Penchala, S. Iannucci, J. Li, S. Rahimi, N. Amiri Golilarz. *Cognitive Firewall: A Proactive, Zero-Trust, Multi-Gate Framework for LLM Safety.* IEEE ICMLA 2026 (accepted). arXiv:2607.01277
- R. Patel, S. Mitra, **M. Guida**, S. Iannucci, S. Mittal, S. Rahimi. *Agentra: A Supervisable Multi-Agent Framework for Enterprise Intrusion Response.* IEEE ICMLA 2026 (accepted). arXiv:2606.18325
- **M. Guida**, S. Iannucci, R. Patel, S. Rahimi, S. Mittal, P. Merialdo. *Shortcut Learning in Early Intrusion Detection: Header Fingerprints Across Four Datasets.* IEEE TPS 2026 (accepted). GitHub: micheleguidaa/early-nids-artifact

PROJECTS

- **Scientific Search Engine (PubMed / arXiv): Retrieval-Augmented IR** — Roma Tre, Data Engineering • 2026
  - Built a multi-entity search engine over biomedical literature on Elasticsearch, indexing papers, figures and tables with BM25 and unified cross-index retrieval; Python pipeline in Docker with Streamlit and CLI interfaces.
  - Evaluated it with an automated framework combining LLM-as-judge and IR metrics (nDCG@5, MAP, Precision@5, MRR) over a 25-query golden set, with error analysis.

- **AI4ESOPP: Energy-Aware Industrial Scheduling** commissioned by mashfrog • 2026

  - Reached zero weighted tardiness (100% on-time) on every evaluation episode while shifting load to cheaper energy windows, with a Maskable PPO agent trained via curriculum learning and fine-tuning.
  - Formulated energy-aware flexible flow-shop scheduling with lot streaming as a Markov Decision Process, jointly minimizing weighted tardiness and time-dependent energy cost, with action masking and wait actions.

- **Explainable Intrusion Detection System (X-IDS)** B.Sc. Thesis • 2024

  - Reached over 97% multiclass accuracy on CIC-IDS-2018 with an explainable IDS based on Self-Organizing Maps, also tested on UNSW-NB15, with XAI visualizations (U-Matrix, class-assignment maps).

EDUCATION

- **Master's Degree in Computer Engineering** Sep 2024 – Jul 2026
  Roma Tre University
  Final grade **110/110 cum laude** • weighted exam average 30.7/30 (Italian scale, honours = 31)
  Key courses: Machine Learning, Deep Learning, Artificial Intelligence, Data Engineering, Software Systems Architecture.

- **Bachelor's Degree in Computer Engineering** Oct 2021 – Jul 2024
  Roma Tre University
  Final grade **110/110 cum laude** • weighted exam average 29.2/30 (Italian scale, honours = 31)
  Key courses: Algorithms & Data Structures, Operating Systems, Networking, Databases, Object-Oriented Programming.

SKILLS

Programming Languages	Python, Java, C, SQL, Bash
ML & Data Science	PyTorch, TensorFlow/Keras, scikit-learn, XGBoost, NumPy, pandas; deep learning (CNNs, RNNs/LSTM, Transformers); reinforcement learning (PPO); feature engineering, model evaluation & explainability
AI / NLP / LLM	large language models (LLMs), NLP, RAG & Graph RAG, agentic / multi-agent systems, LangChain / LangGraph, LLM APIs (OpenAI, Anthropic, Gemini), local LLMs (Ollama), prompt engineering, LLM safety & guardrails, LLM-as-judge evaluation
Data & Retrieval	Elasticsearch, BM25 & embeddings, information retrieval, IR evaluation (nDCG, MAP, MRR), ETL & data integration, knowledge graphs, relational databases
DevOps & Tools	Docker, Git, CI/CD, Linux, REST APIs, reproducible ML pipelines, Proxmox / VMware
Security	intrusion detection (IDS), network / flow analysis, anomaly detection, TCP/IP, MITRE ATT&CK, MITRE D3FEND, NIST CSF
Languages	Italian (Native) and English (C1 – Advanced)

INTERESTS

Homelabbing as a hands-on, continuous-learning lab: self-hosted services in Docker containers on a Proxmox / VMware stack running on Linux, and building with GenAI coding agents (Claude Code, Codex). I follow AI research and take part in the IT community (Linux Day, tech fairs). Beyond tech, I enjoy travel, padel and pickleball.

I authorise the processing of my personal data contained in this CV pursuant to art. 13 of Regulation (EU) 2016/679 (GDPR) and Italian Legislative Decree 196/2003, as amended by Legislative Decree 101/2018.